PSY 5790: Applied Univariate and Bivariate/Bivariable Statistics

Correlation and Simple Linear Regression Demo: Categorical Predictor

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Required Libraries

```
library(tidyverse)
library(dplyr)
library(kableExtra)
library(psych)
library(jmv)
library(lm.beta)
library(skimr)
library(broom)
library(moderndive)
library(gapminder)
```

Next, load in the data stored in `gapminder` package

```
gapminder2007 <- gapminder %>%
  filter(year == 2007) %>%
  select(country, continent, lifeExp, gdpPercap)
```

Exploratory Data Analysis

1. Observe raw data
2. Compute descriptive statistics
3. Visualize data

Observe raw data

```
head(gapminder2007)
```

```
## # A tibble: 6 × 4
##   country     continent lifeExp gdpPercap
##   <fct>       <fct>       <dbl>     <dbl>
## 1 Afghanistan Asia         43.8      975.
## 2 Albania     Europe       76.4     5937.
## 3 Algeria     Africa       72.3     6223.
## 4 Angola      Africa       42.7     4797.
## 5 Argentina   Americas     75.3    12779.
## 6 Australia   Oceania      81.2    34435.
```

```
#glimpse(gapminder2007)
#View(gapminder2007)
```

Compute descriptive statistics

```
# Can use any of these but my preference is the `describe()` function
describe(gapminder2007)
```

```
##            vars   n     mean       sd  median trimmed     mad    min      max
## country*      1 142    71.50    41.14   71.50   71.50   52.63   1.00   142.00
## continent*    2 142     2.33     1.21    2.00    2.27    1.48   1.00     5.00
## lifeExp       3 142    67.01    12.07   71.94   68.11   11.01  39.61    82.60
## gdpPercap     4 142 11680.07 12859.94 6124.37 9614.17 7522.41 277.55 49357.19
##               range  skew kurtosis      se
## country*     141.00  0.00    -1.23    3.45
## continent*     4.00  0.25    -1.36    0.10
## lifeExp       42.99 -0.67    -0.87    1.01
## gdpPercap  49079.64  1.20     0.25 1079.18
```

```
## These alternatives give you many descriptive stats but nothing about normality
# skim(gapminder2007)
# summary(gapminder2007)
# skim_without_charts(gapminder2007)

## Don't forget correlations --- Can make this work for character/factor variables!!
corrMatrix(gapminder2007,
           kendall = TRUE)
```

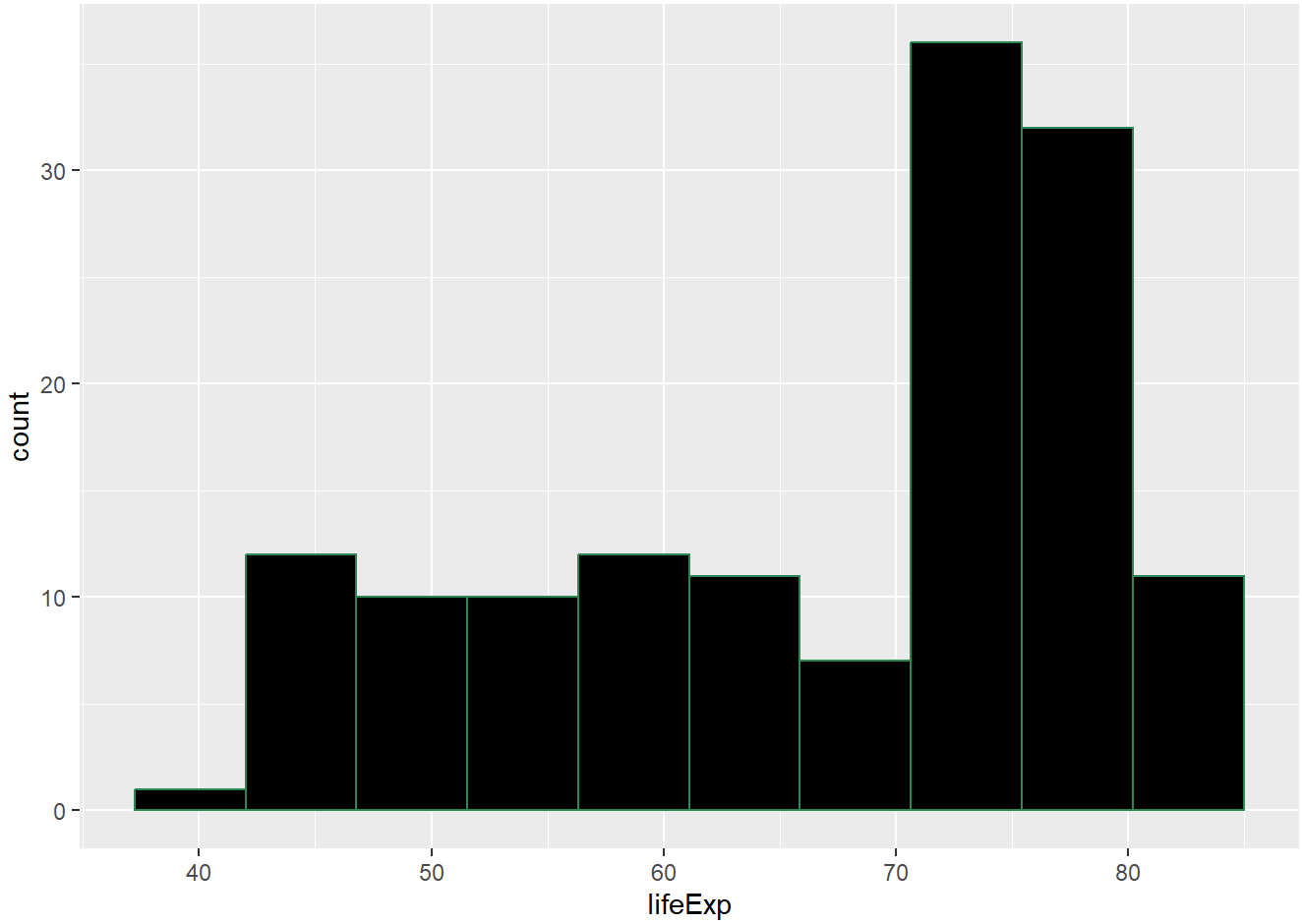
```
##
##  CORRELATION MATRIX
##
##  Correlation Matrix
##  ──────────────────────────────────────────────────────────────────────────────
##                                   country      continent     lifeExp       gdpPercap
##  ──────────────────────────────────────────────────────────────────────────────
##    country      Pearson's r               —
##                 df                        —
##                 p-value                   —
##                 Kendall's Tau B           —
##                 p-value                   —
##
##    continent    Pearson's r             NaN             —
##                 df                      140             —
##                 p-value                 NaN             —
##                 Kendall's Tau B    0.0186791            —
##                 p-value            0.7673888            —
##
##    lifeExp      Pearson's r             NaN           NaN             —
##                 df                      140           140             —
##                 p-value                 NaN           NaN             —
##                 Kendall's Tau B    0.0218759     0.5927423            —
##                 p-value            0.6993055    < .0000001            —
##
##    gdpPercap    Pearson's r             NaN           NaN     0.6786624            —
##                 df                      140           140           140            —
##                 p-value                 NaN           NaN    < .0000001            —
##                 Kendall's Tau B    0.0114874     0.5228988     0.6779543            —
##                 p-value            0.8392682    < .0000001    < .0000001            —
##  ──────────────────────────────────────────────────────────────────────────────
```

```
## By default corrMatrix() calculates Pearson's r BUT adding additional arguments
## Can provide subsequent assessments that do not require continuous variables
## spearman = TRUE --- allows for correlations among categorical variables
## kendall = TRUE --- allows for correlations among continuous/categorical variables
```

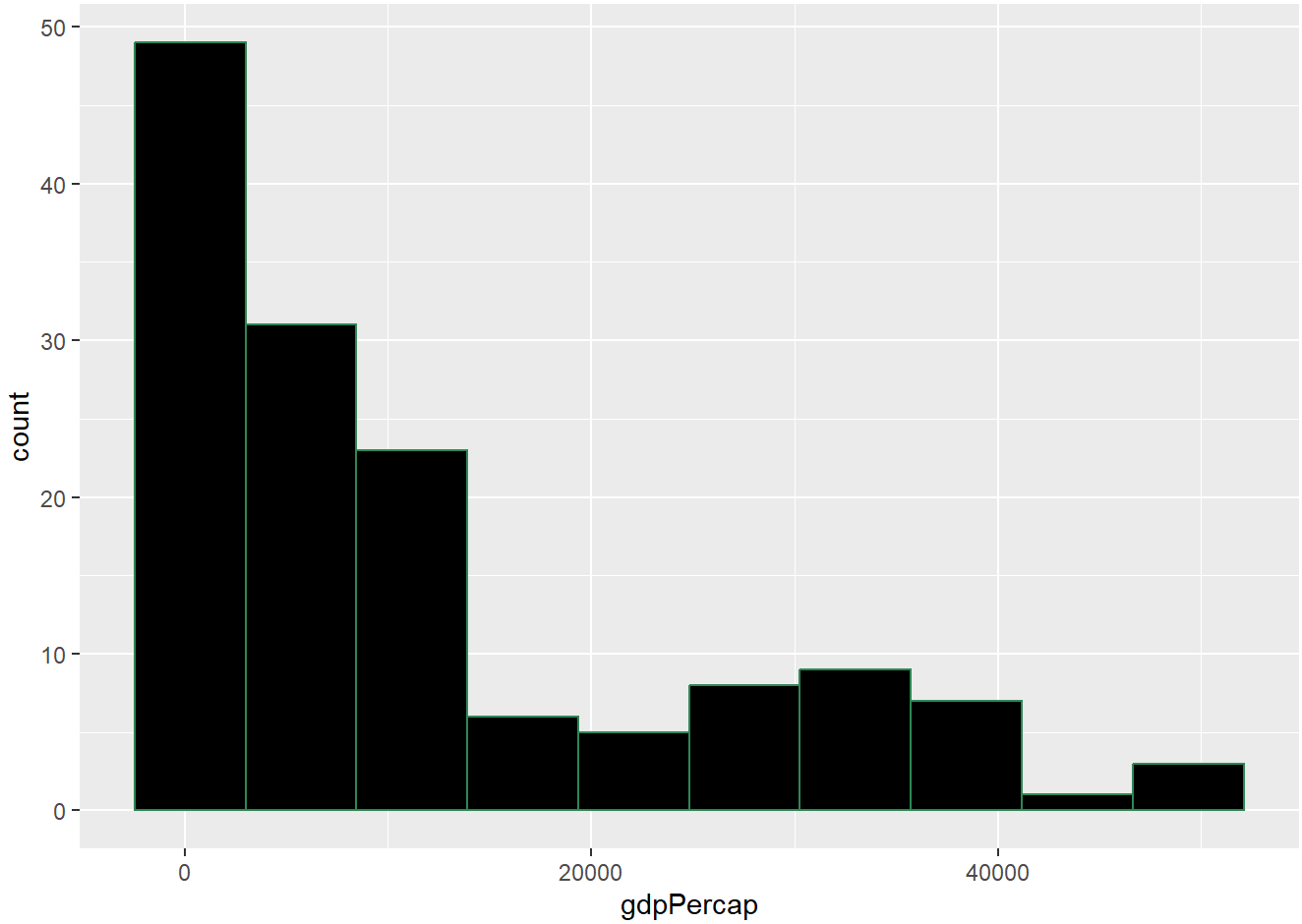
Visualize the data

```
## Histograms

ggplot(data = gapminder2007,
       mapping = aes(x = lifeExp)) +
  geom_histogram(bins = 10,
                 color = "seagreen",
                 fill="black")
```

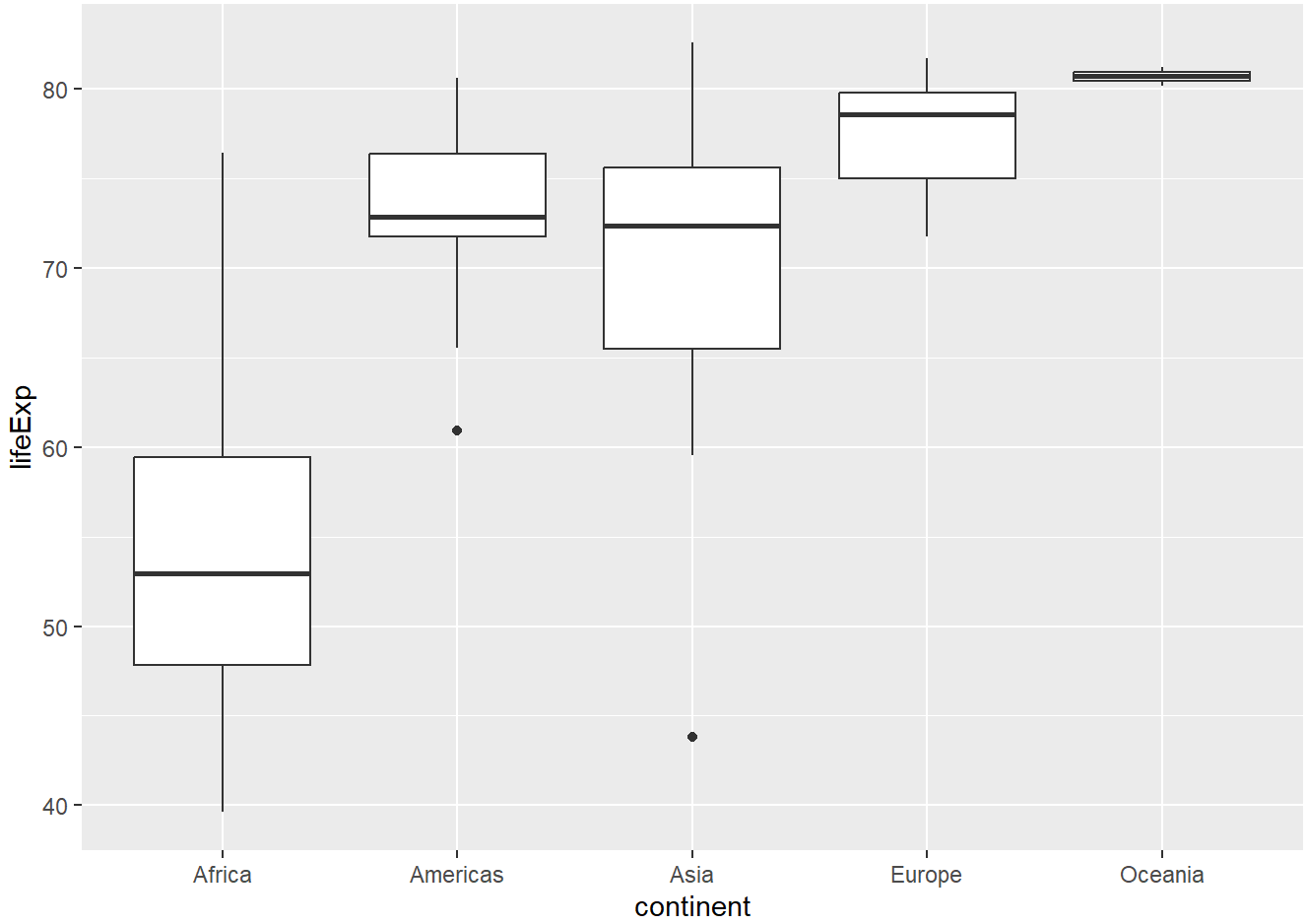



```
ggplot(data = gapminder2007,
       mapping = aes(x = gdpPercap)) +
  geom_histogram(bins = 10,
                 color = "seagreen",
                 fill="black")
```

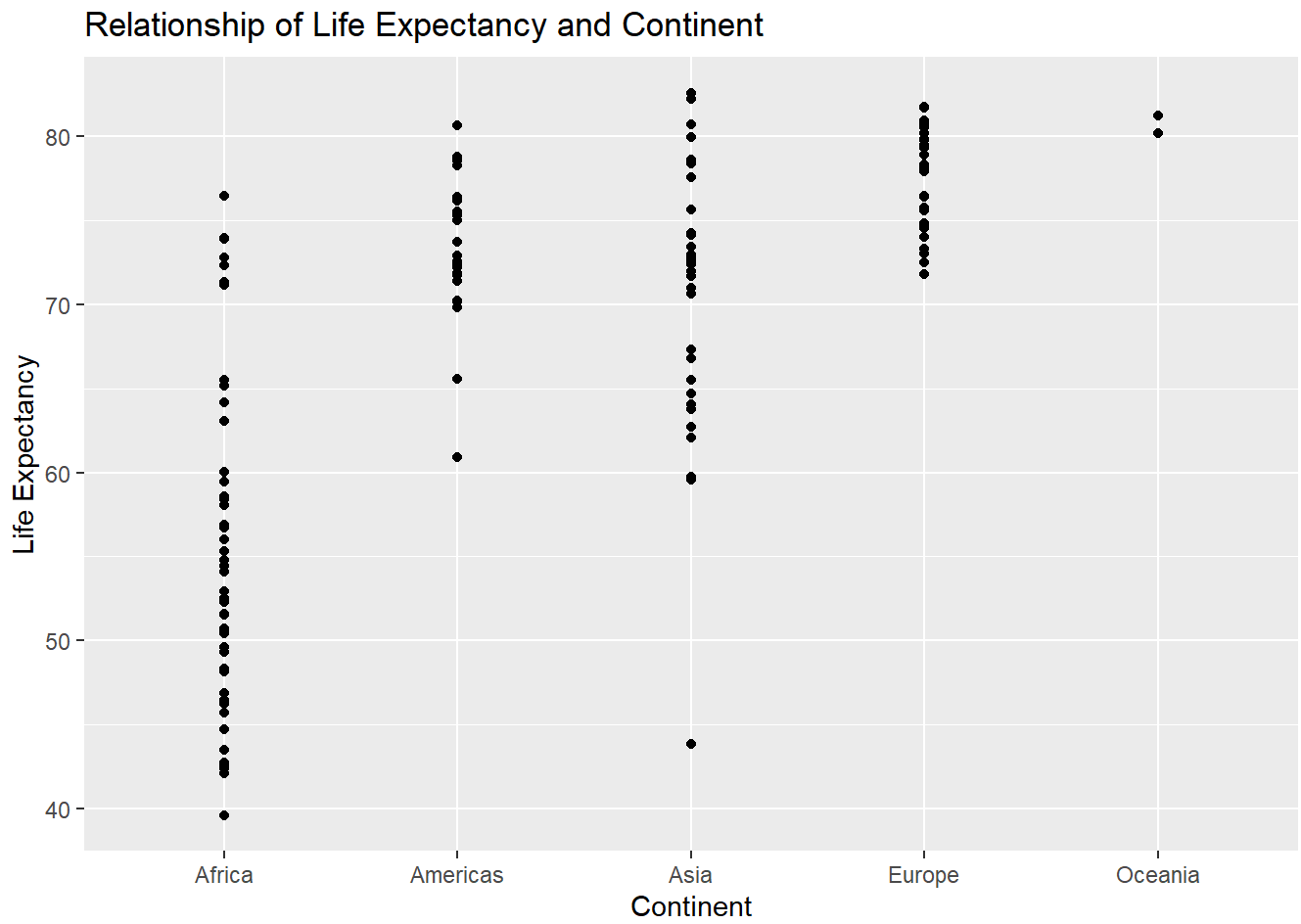



```
## Side-by-Side Boxplot

ggplot(data = gapminder2007,
       mapping = aes(y = lifeExp,
                     x = continent)) +
  geom_boxplot()
```




```
## Scatterplot
# Visualization will not generate linear regression line so this has been removed from this visu
alization
ggplot(gapminder2007,
       aes(x = continent,
           y = lifeExp)) +
  geom_point() +
  labs(x = "Continent",
       y = "Life Expectancy",
       title = "Relationship of Life Expectancy and Continent")
```



Now lets run our simple linear regression (`lm()` workflow + `broom` package)

```
lifeExp_model <- lm(lifeExp ~ continent, data = gapminder2007)

# Get regression table using `tidy()`:
tidy(lifeExp_model, conf.int=TRUE)
```

```
## # A tibble: 5 × 7
##   term              estimate std.error statistic  p.value conf.low conf.high
##   <chr>                <dbl>     <dbl>     <dbl>    <dbl>    <dbl>     <dbl>
## 1 (Intercept)           54.8      1.03     53.4  1.23e-93     52.8      56.8
## 2 continentAmericas     18.8      1.80     10.4  3.74e-19     15.2      22.4
## 3 continentAsia         15.9      1.65      9.67 3.41e-17     12.7      19.2
## 4 continentEurope       22.8      1.70     13.5  7.10e-27     19.5      26.2
## 5 continentOceania      25.9      5.33      4.86 3.12e- 6     15.4      36.4
```

```
# Get model fit table using `glance()`:
glance(lifeExp_model)
```

```
## # A tibble: 1 × 12
##   r.squared adj.r.squared sigma statistic  p.value    df logLik   AIC   BIC
##       <dbl>         <dbl> <dbl>     <dbl>    <dbl> <dbl>  <dbl> <dbl> <dbl>
## 1     0.635         0.625  7.39      59.7 4.22e-29     4  -483.  978.  996.
## # ℹ 3 more variables: deviance <dbl>, df.residual <int>, nobs <int>
```

Alternatively we can use `linReg()` from the `jmv` package

```
linReg(gapminder2007,
       dep=lifeExp,
       factors=continent,
       blocks=list(
         list('continent')),
       anova=TRUE,
       stdEst=TRUE,
       ci=TRUE,
       ciStdEst=TRUE)
```

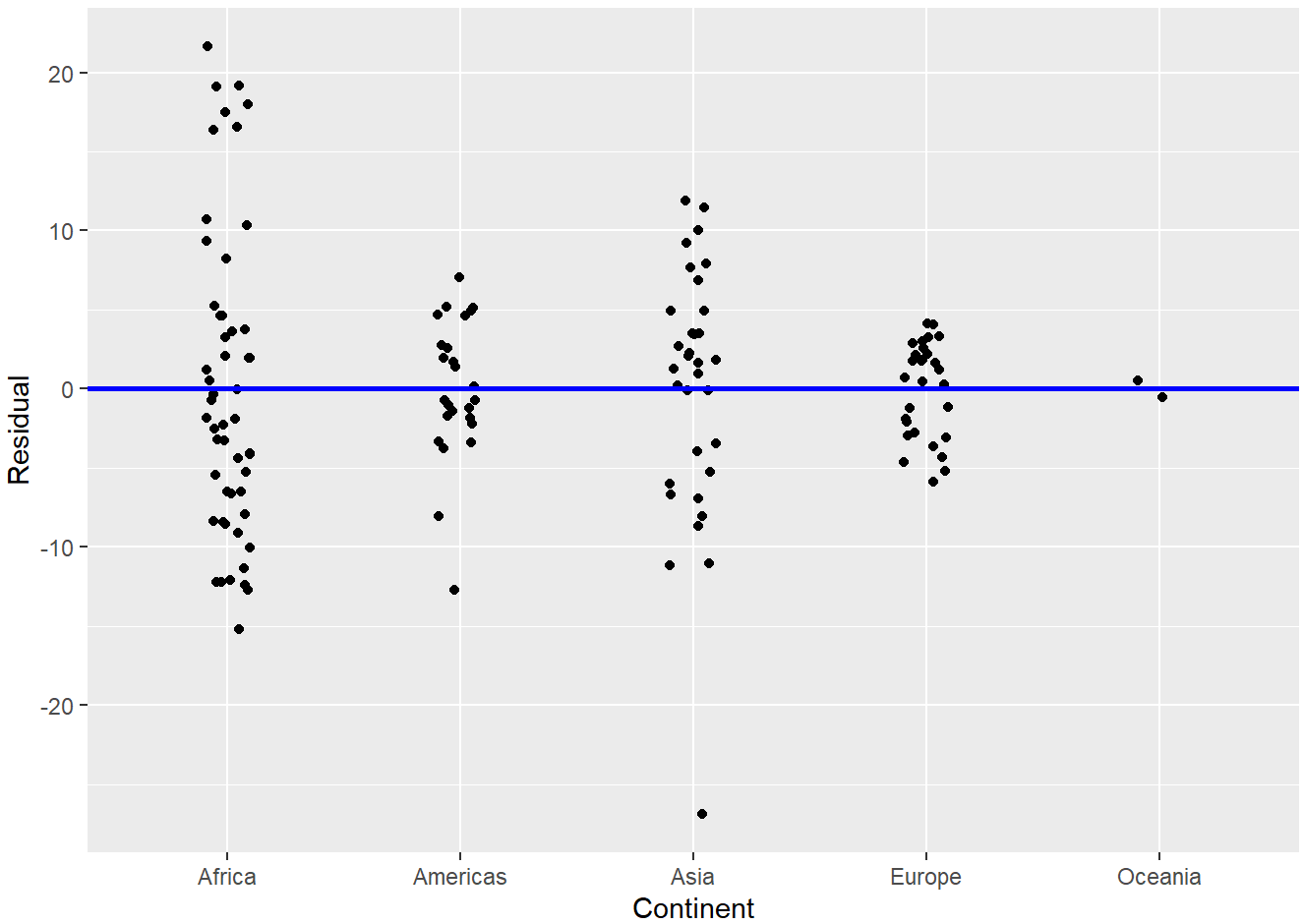
```
##
##  LINEAR REGRESSION
##
##  The specified reference level was not found for the following
##  variable(s): 'continent'. Defaulting to the first available level. To
##  use a custom reference level, ensure the defined reference level is
##  available in the data.
##
##  Model Fit Measures
##  ─────────────────────────────────────
##    Model    R            R²
##  ─────────────────────────────────────
##        1    0.7971817    0.6354987
##  ─────────────────────────────────────
##    Note. Models estimated using
##    sample size of N=142
##
##
##  MODEL SPECIFIC RESULTS
##
##  MODEL 1
##
##  Omnibus ANOVA Test
##  ──────────────────────────────────────────────────────────────────────────────
##                 Sum of Squares    df     Mean Square    F           p
##  ──────────────────────────────────────────────────────────────────────────────
##    continent         13060.676      4     3265.16908    59.71400    < .0000001
##    Residuals          7491.177    137       54.68012
##  ──────────────────────────────────────────────────────────────────────────────
##    Note. Type 3 sum of squares
##
##
##  Model Coefficients - lifeExp
##  ──────────────────────────────────────────────────────────────────────────────
```

```
_____________________________________________
##    ª Represents reference level
```

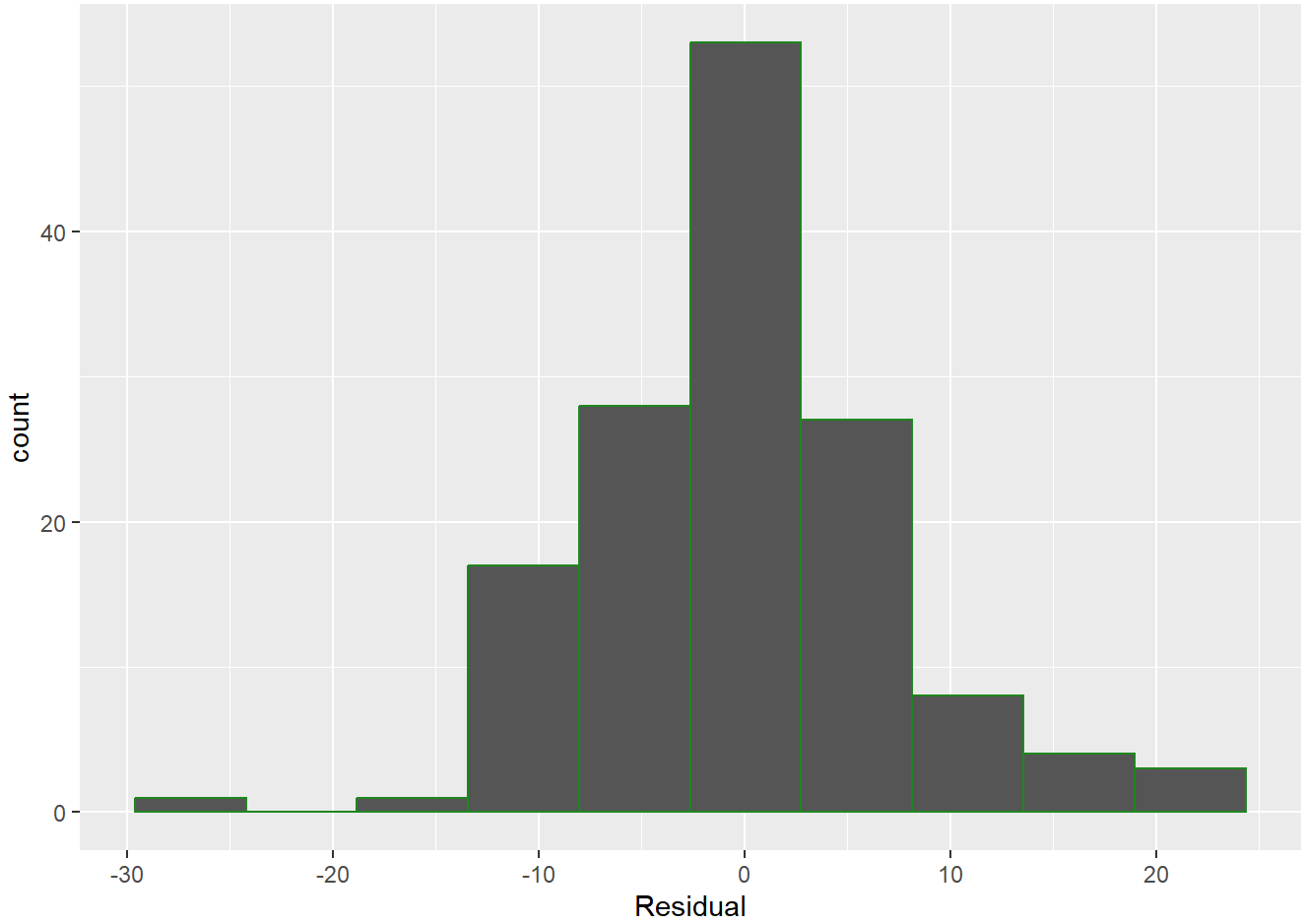
Lastly we should conduct a residual analysis

```
regression_points <- get_regression_points(lifeExp_model)

# Scatterplot of Explanatory Variable along x-axis and Residuals along y-axis
ggplot(regression_points,
       aes(x = continent, y = residual)) +
  geom_jitter(width = 0.1) +
  labs(x = "Continent", y = "Residual") +
  geom_hline(yintercept = 0, col = "blue", size = 1)
```




```
# Additionally we want to look at the distribution of residuals
ggplot(regression_points,
       aes(x = residual)) +
  geom_histogram(bins = 10, color = "forestgreen") +
  labs(x = "Residual")
```




```
describe(regression_points$residual)
```

```
##    vars   n mean   sd median trimmed  mad   min   max range skew kurtosis   se
## X1    1 142    0 7.29   0.26   -0.21 5.45 -26.9 21.64 48.54 0.16     1.44 0.61
```